Multi-Turn AI Chatbot with LLaMA 3

Project Domain / Category

Artificial Intelligence / Natural Language Processing / Conversational AI

Abstract / Introduction

This project focuses on the development of a multi-turn AI chatbot using a local **LLaMA 3** that enables users to engage in long, dynamic conversations across multiple domains such as machine learning, healthcare, power systems, and e-commerce etc. The chatbot processes user queries in real time without prompt engineering and supports continuous dialogue through a structured session-based system.

In addition to core chatbot functionality, the system incorporates an intelligent context management mechanism that detects topic changes and dynamically manages the context window to improve response relevance. A mandatory feedback mechanism is integrated where users rate each response based on correctness, quality, and length. All interactions are stored in a structured database for further analysis.

The system performs multi-dimensional analytics including topic-wise performance, response accuracy, positional analysis (start, middle, and end of sessions), response time evaluation, and correctness per reply. These insights are presented through interactive tables and graphical dashboards, making the system a comprehensive AI-driven conversational platform combined with advanced behavioral and performance analysis capabilities.

Functional Requirements:

PHASE 1: Chatbot System Development (Frontend + Backend Setup)

FR1: System Initialization

- The system shall initialize all core modules at startup:
  - LLaMA 3 local model loading
  - Backend server initialization
  - Database connection setup
  - Analytics module initialization

FR2: Frontend Chat Interface Development

- The system shall provide a web-based chat UI including:
  - Google OAuth login page
  - Chat interface (user input + response display)
  - Feedback panel (rating, correctness, length type)

FR3: Backend Server Development

- The system shall implement a backend using Flask/FastAPI.
- The backend shall handle:
  - User authentication
  - Session creation and management
  - Message routing to model
  - Data storage operations

FR4: LLaMA 3 Model Integration

- The system shall integrate **LLaMA 3** as the core chatbot engine.
- The system shall process user queries locally through the model.

FR5: Service Connectivity

- The system shall connect:
  - Frontend UI
  - Backend API
  - LLaMA 3 model inference engine
  - Database
  - Analytics module

PHASE 2: Chat Execution + Data Storage System

FR6: User Authentication

- The system shall allow login via Google OAuth.
- The system shall create or retrieve user profiles after login.

FR7: Session Management

- The system shall generate a unique session ID for each chat session.
- Each session shall store conversation state independently.

FR8: Multi-Turn Chat Functionality

- The system shall allow continuous conversation within a session.
- The system shall support multiple topics in a single session.

FR9: LLaMA 3 Response Generation

- The system shall send user queries to LLaMA 3 model, receive and display generated responses in real time.

FR10: Response Time Tracking

- The system shall calculate response time using request and response timestamps.

FR11: Real-Time Data Storage

- The system shall store every interaction:
  - User message
  - Model response
  - Session ID
  - Timestamp
  - Topic label

FR12: Mandatory Feedback System

- The system shall freeze input after each response.
- The system shall require user feedback before next message:
  - Rating (1–4)
  - Correctness (Correct / Partial / Incorrect)
  - Length type (Short / To the Point / Lengthy)

FR13: Input Control Mechanism

- The system shall block new input until feedback is submitted.
- The system shall unlock chat after feedback completion.

PHASE 3: Analytics System

FR14: Data Retrieval for Analytics

- The system shall fetch stored session data from the database for analysis.

FR15: Topic Classification

- The system shall classify messages into domains:
  - Machine Learning
  - Deep Learning
  - Healthcare AI
  - Power Systems
  - E-commerce AI

FR16: Session Segmentation

- The system shall divide sessions into:
    - Start phase
    - Middle phase
    - End phase

FR17: Core Metric Computation

- The system shall compute:
    - Accuracy per reply
    - Accuracy per topic
    - Accuracy per phase
    - Response time statistics
    - Rating averages
    - Length distribution

FR18: Correctness Evaluation

- The system shall calculate:
    - Correct replies count
    - Partially correct replies count
    - Incorrect replies count
    - Overall correctness percentage

FR19: Table Generation

- The system shall generate analytical tables:
    - Topic vs accuracy
    - Reply correctness table
    - Phase-wise accuracy
    - Response time summary
    - Rating distribution
    - Length preference

FR20: Graph Generation

- The system shall generate visual graphs:
    - Topic distribution (pie chart)
    - Accuracy per topic (bar chart)
    - Accuracy vs phase (line graph)
    - Correctness per reply (bar chart)
    - Response time trend (line graph)
    - Rating distribution (bar chart)
    - Length preference (pie chart)

FR21: Dashboard Display

- The system shall display analytics dashboard including:
  - Tables
  - Graphs
  - Session-level insights
  - Topic-based performance

FR22: Session Termination

- The system shall end the session and store the final analytics summary permanently.

Tools:

- **Python** – Free and open-source programming language used for backend development, chatbot logic, and analytics.
- **LLaMA 3** – Free/open-source large language model used as the core chatbot engine for generating responses locally.
- **Flask / FastAPI** – Free and open-source Python web frameworks used to build backend APIs and manage chatbot requests.
- **HTML, CSS, JavaScript** – Standard web technologies for building the chatbot user interface.
- **MongoDB / Firebase** – Free-tier cloud databases used for storing chat logs, sessions, and feedback data.
- **pandas** – Open-source Python library used for data processing and analytics.
- **matplotlib / seaborn / plotly** – Open-source libraries used for generating graphs and visual analytics.
- **scikit-learn** – Open-source machine learning library used for topic classification and analytics support.
- **Ollama (optional)** – Free tool used to run LLaMA 3 locally and serve it as an API.
- **Google OAuth 2.0** – Free authentication system used for secure Gmail login integration.
- **VS Code / Jupyter Notebook** – Free development environments used for coding, testing, and experimentation.
- **Git & GitHub** – Free version control system used for project management and collaboration.

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